Functions & Graphs

Section 1: Function Evaluation & Inverses

1 If $f(x) = 3x^2 - 4x + 7$, what is the value of $f(-2)$?

a. 11

b. 21

c. 27

d. 31

2 Given the linear function $f(x) = \frac{4x - 7}{3}$, what is the equation of the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x + 7}{4}$

b. $f^{-1}(x) = \frac{3x - 7}{4}$

c. $f^{-1}(x) = \frac{4x + 3}{7}$

d. $f^{-1}(x) = \frac{7x + 3}{4}$

3 If $g(x) = \sqrt{2x + 10}$, what is the domain of $g(x)$ in the set of real numbers?

a. $x \geq -5$

b. $x > -5$

c. $x \geq 0$

d. $x \leq -5$

Section 2: Composite Functions & Domain/Range

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4 Let $f(x) = 2x + 5$ and $g(x) = x^2 - 3$. What is the value of $(g \circ f)(1)$?

a. 11

b. 46

c. 49

d. 52

5 If $h(x) = 3x - 1$ and $(h \circ k)(x) = 6x + 8$, which of the following defines $k(x)$?

a. $k(x) = 2x + 3$

b. $k(x) = 2x + 9$

c. $k(x) = 3x + 4$

d. $k(x) = 2x + 2$

6 What is the range of the quadratic function $f(x) = -(x - 4)^2 + 9$?

a. $y \leq 9$

b. $y \geq 9$

c. $y \leq 4$

d. All real numbers

Section 3: Graph Transformations & Symmetries

7 The graph of $y = f(x)$ is shifted 3 units to the right and 5 units down. What is the equation of the transformed graph?

a. $y = f(x - 3) - 5$

b. $y = f(x + 3) - 5$

c. $y = f(x - 3) + 5$

d. $y = f(x + 3) + 5$

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8 Which of the following functions is an even function, satisfying $f(-x) = f(x)$ for all real x?

a. $f(x) = x^3 - x$

b. $f(x) = 4x^2 + 7$

c. $f(x) = 2x + 1$

d. $f(x) = x^4 - 3x$

9 What is the horizontal asymptote of the rational function $f(x) = \frac{6x^2 - 5x + 1}{2x^2 + 3x - 4}$?

a. $y = 0$

b. $y = 3$

c. $y = 2$

d. No horizontal asymptote

Section 4: Piecewise & Special Functions

10 A function is defined piecewise by $f(x) = 2x + 3$ for $x < 1$, and $f(x) = x^2 - 1$ for $x \geq 1$. What is the value of $f(-2) + f(3)$?

a. 7

b. 8

c. 9

d. 11

11 If $[x]$ denotes the greatest integer function (floor function), what is the value of $[3.8] - [-2.4]$?

a. 5

b. 6

c. 7

d. 1

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12 For what value of c is the piecewise function $f(x) = cx + 4$ for $x \leq 2$, and $f(x) = 3x^2 - 2$ for $x > 2$ continuous at $x = 2$?

a. 2

b. 3

c. 4

d. 5

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